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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program/s: B Tech

Year: I Semester: II

Stream/s: Computer Science and Business Systems

Subject: Data Structures and Algorithms

Time: 3 hrs. (11:00am to 2:00 pm)

Date: 12 / 05 / 2023

No. of Pages: 02

Marks: 100

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1_ is compulsory.
2) Out of remaining questions, attempt any _4_ questions.
3) **In all _5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	
CO- 2; BL- L	a.	What is doubly linked list? Explain with suitable example.	[05]
CO-1; BL- L	b.	What is the significance of abstract data type? Give suitable example.	[05]
CO- 1; BL-M	c.	Explain the factors contributed to measure the performance of an algorithm	[05]
CO- 4; BL-H	d.	Write algorithm / program to perform linear searching with its best, average and worst case analysis.	[05]
Q2 CO-2; BL-M	a	What is the significance of recursion over looping? Explain the use of implicit stack in recursion with suitable example and appropriate steps.	[10]
CO-2; BL-H	b	Values are entered in sequence based on user's choice in appropriate data structure (number of values are not known initially). Perform insert and delete operation from the middle of the data structure with the help of suitable algorithms.	[10]
Q3 CO-4; BL-M	a	Apply and explain binary searching algorithm on the following dataset with best, average and worst case time complexity analysis. 10, 20, 30, 40, 50, 50, 60, 70, 80.	[10]

CO-2; BL-H	b	Convert and evaluate infixed expression: a + (b – c) * (d + (e * 2)) using stack. Write all the necessary steps used during the conversion from infixed to postfixed expression.	[10]
Q4 CO-2; BL-M	a	Write algorithm / program to read 10 elements in static data structure. Perform one delete and one insert operation at appropriate location entered by user. Display the final content of the data structure.	[10]
CO-3; BL-L	b	Explain the significance of all the binary tree traversal algorithms with suitable example.	[10]
Q5 CO-4; BL-L	a	Explain merge sort algorithm with suitable example. Discuss its worst case time complexity.	[10]
CO-3; BL-H	b	Consider the following message and encode it using Huffman encoding method: AAABBCBBACCCBBDDDDEEAABBB Also draw and explain the use of binary tree in encoding the message.	[10]
Q6 CO-3; BL-L	a	What is minimum spanning tree (MST)? Explain the working principle of MST with suitable example.	[10]
CO-3; BL-M	b	Given a sequence of elements: 45, 23, 56, 78, 12, 32, 80, 66, 6, 69, 15. Apply and explain the algorithm / function to generate the binary search tree for this given sequence.	[10]
Q7 CO-3; BL-M	a	Draw the graph with at least 8 nodes and 12 edges as an example. Apply breadth first search method to traverse this graph and display the final sequence of traversal.	[10]
CO-2; BL-L	b	How do you do insertion of node at the beginning and at the end of the singly linked list. Explain with suitable algorithm / program.	[10]